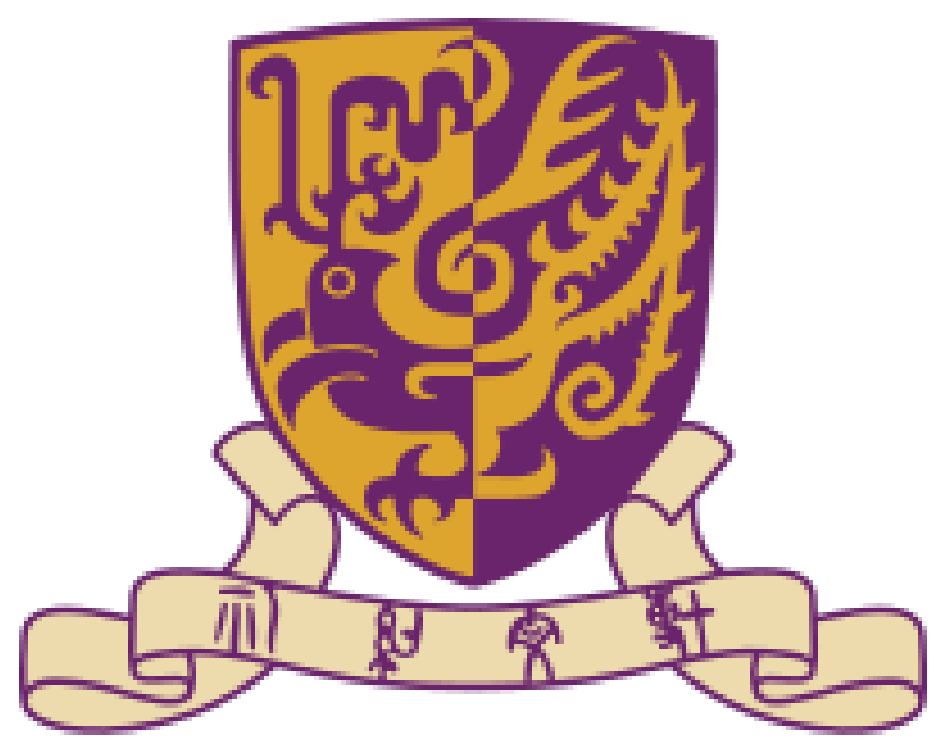




RW1: Design a short-range remote methane detector

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I. Introduction (or any background information)

• Background (or an equivalent name for this sub-section)

1. Methane (CH_4) is an invisible, odourless and highly flammable green house gas.
2. By the Beer law, absorbance is proportional to the gas concentration and optical path length.

• Problem Statement (or any problem to be solved)

1. The traditional method are slow and limited to the hidden leakage.
2. Lack of convenient solution to detect methane with accurate data analysis.
3. Fast, reliable data analysis and graphing are required for real-life industrial safety applications
4. And we are able to do some extension calculation and analysis base on the graphing and data observe from this system.

II. Materials / Models and Methods

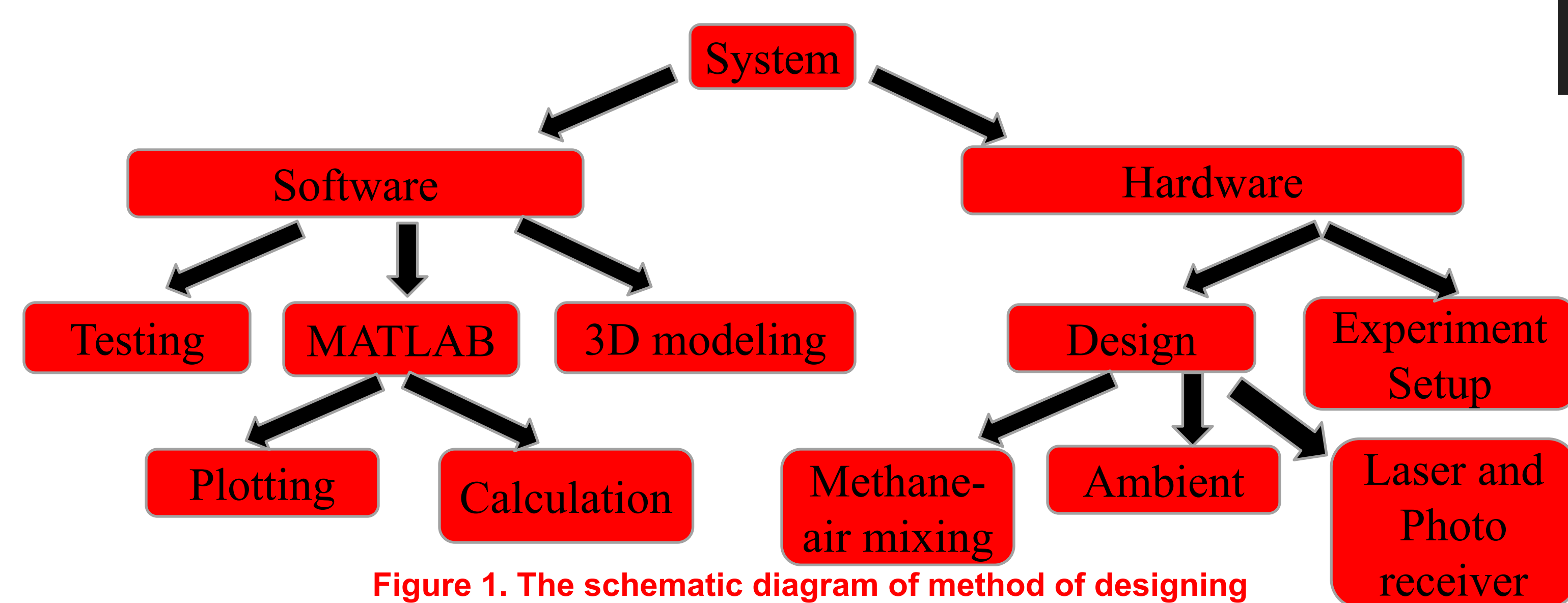


Figure 1. The schematic diagram of method of designing

i. sub-section for the materials, models and methods

$$T_v = \left(\frac{I}{I_0} \right)_v = \exp(-k_v L) - \text{Beer Law}$$

$$\text{Absorbance } A_e = \ln \left(\frac{I_0}{I} \right) = -\ln(T_v) = k_v L = \alpha C L$$

Where,

T_v = Transmittance

I, I_0 = Transmitted, Incident Intensity

k_v = Absorption Coefficient

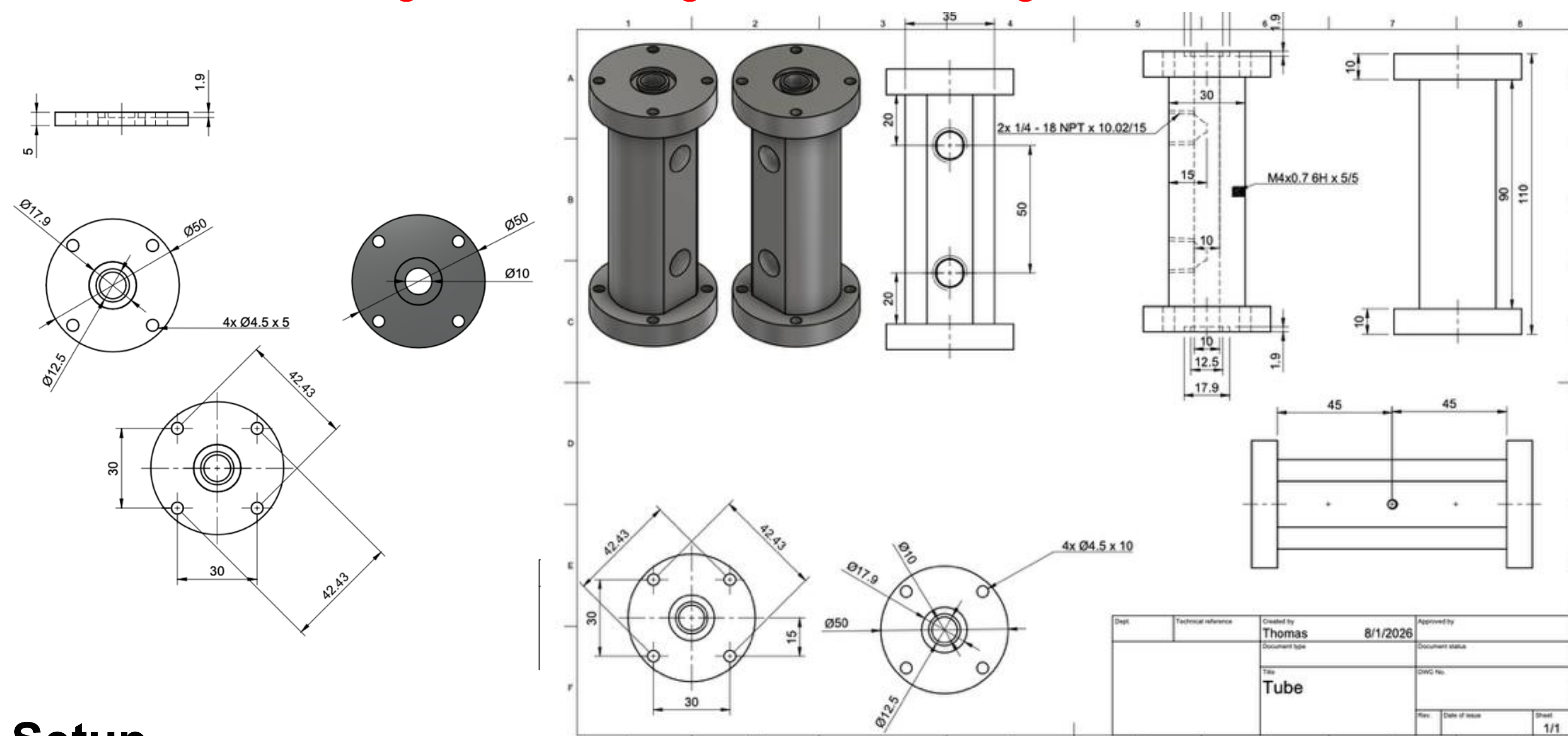
L = Path Length

III. Results and Discussion

• Design

- Total length of 14.16cm length system, using 2 25x6mm wedge window with $0.5^\circ \pm 0.1667^\circ$ tiny angle.

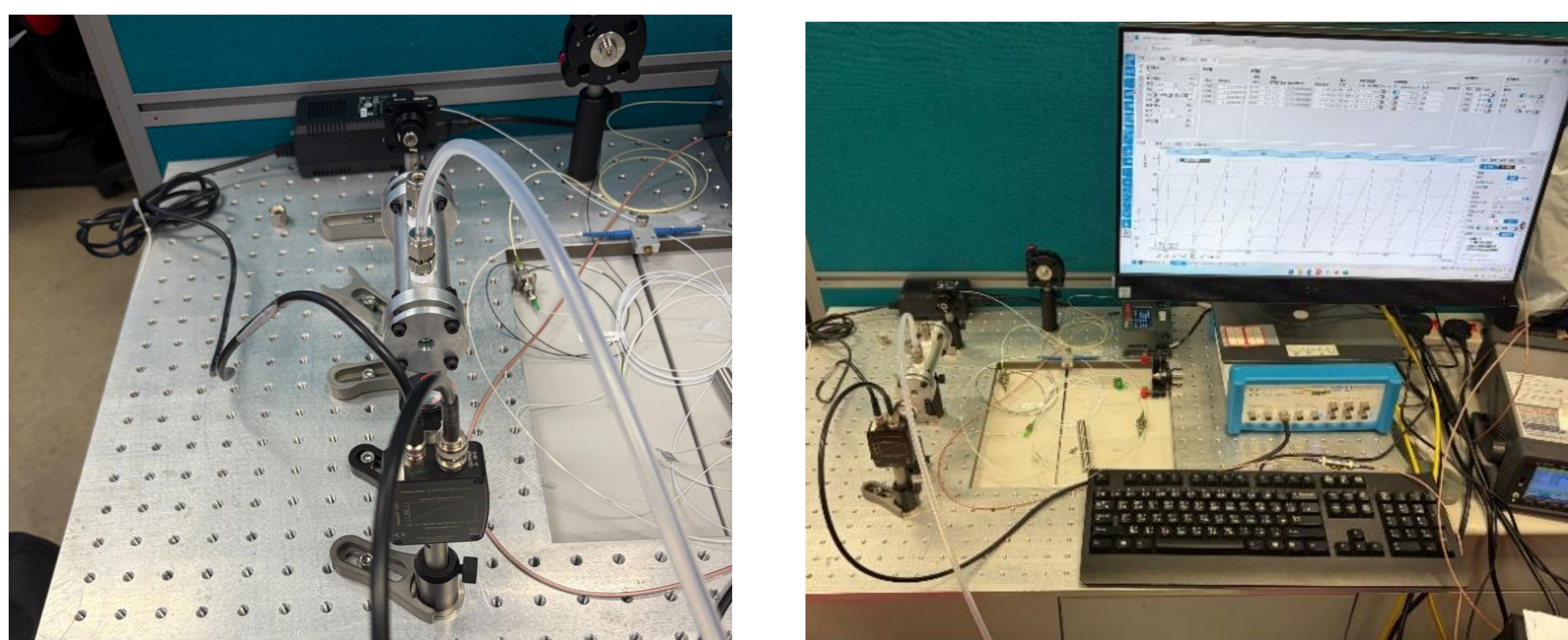
Figure 1,2 The design schematic of the gas cell



• Setup

- The experiment setup for methane-air mixture testing and observe.

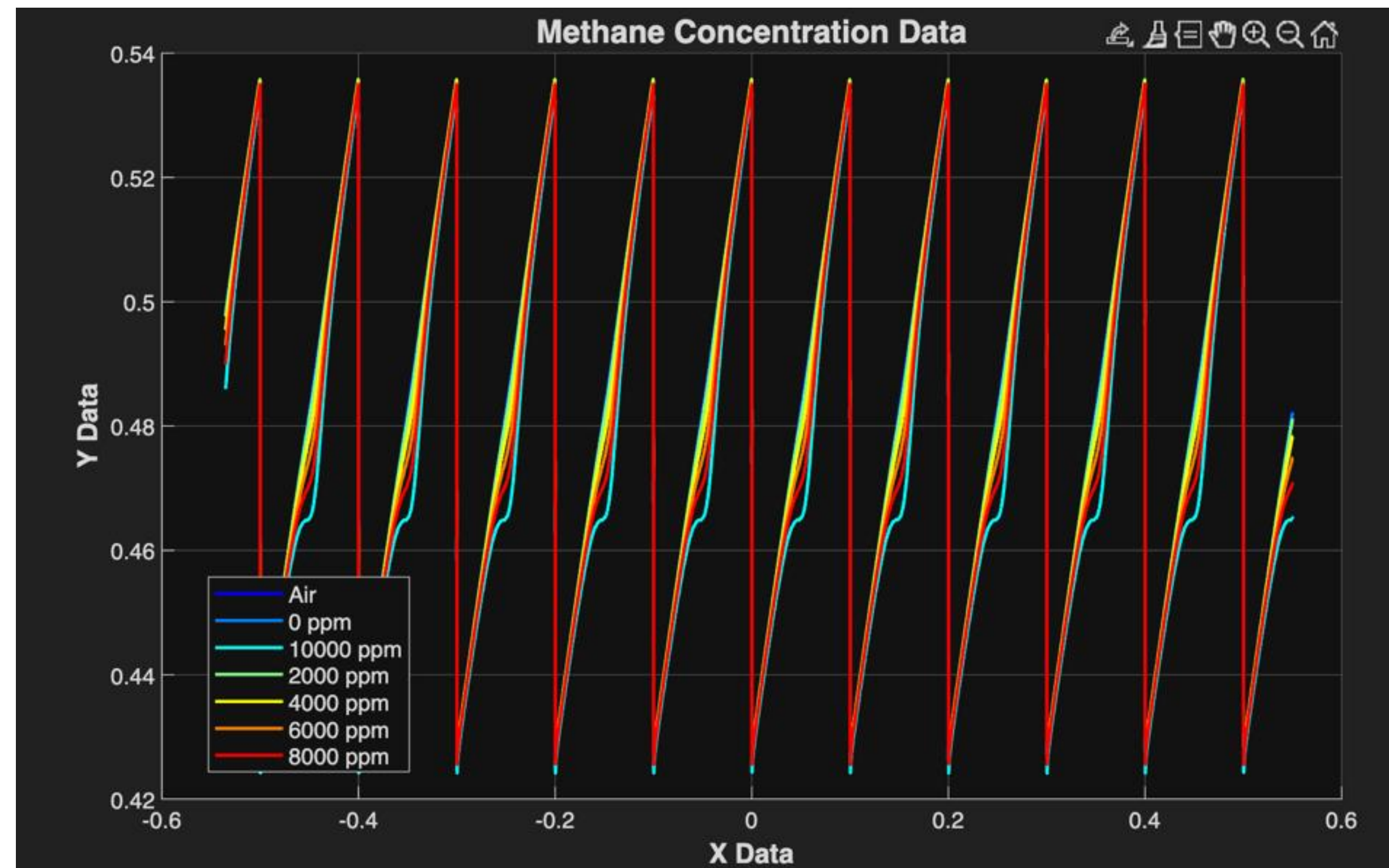
Figure 3. The experimental setup of the TDLAS system



III. Results and Discussion (cont.)

i. Sub-section for the discussion

Figure 4. Mixture data from different concentration



Low concentration (0-2000) ppm:

- The curve is a basically linear form

Mid concentration (4000-8000) ppm:

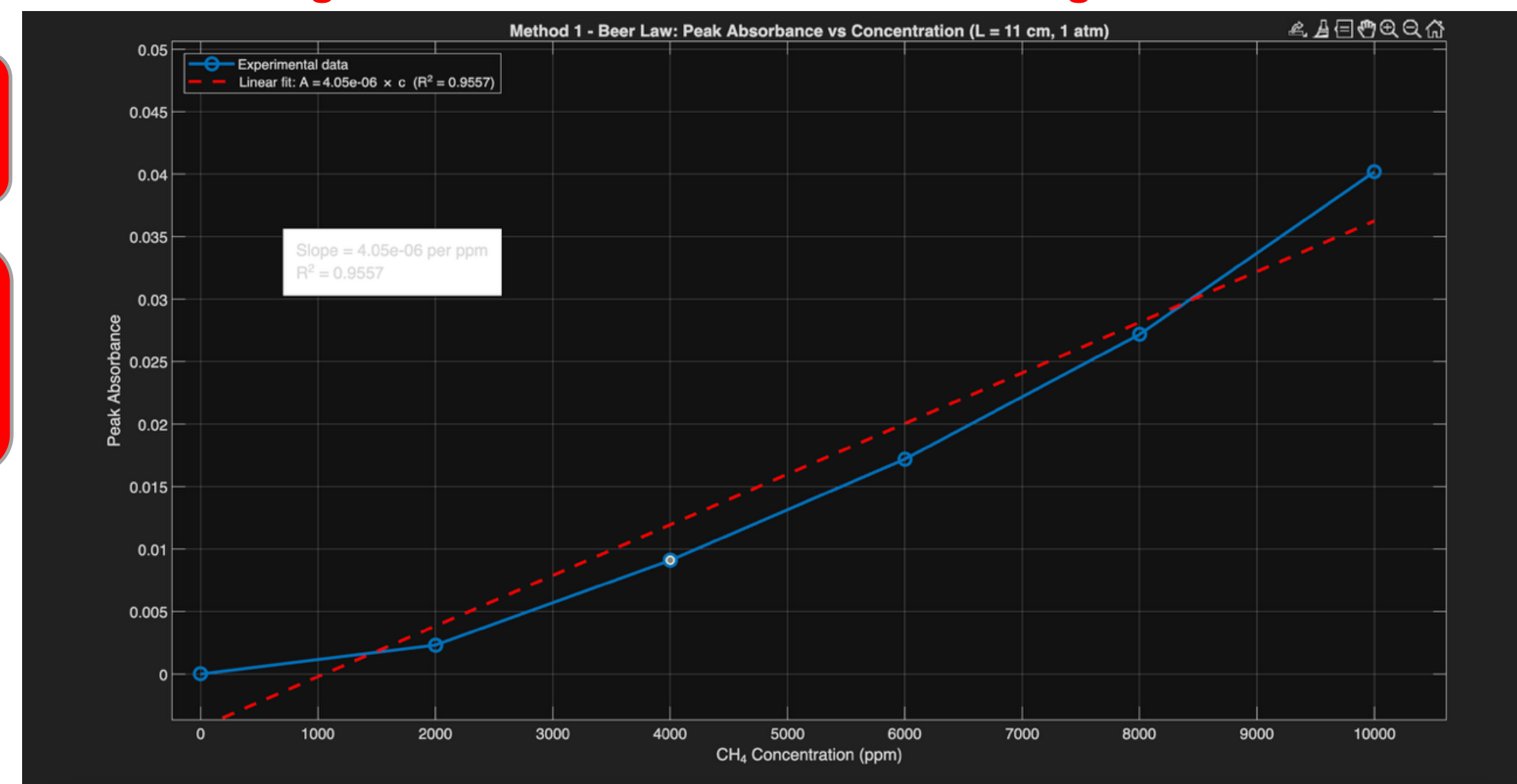
- The bubble shape starts to appear in the curve and affecting the linear shape

High concentration (10000) ppm:

- The saturation bubble becomes most significant

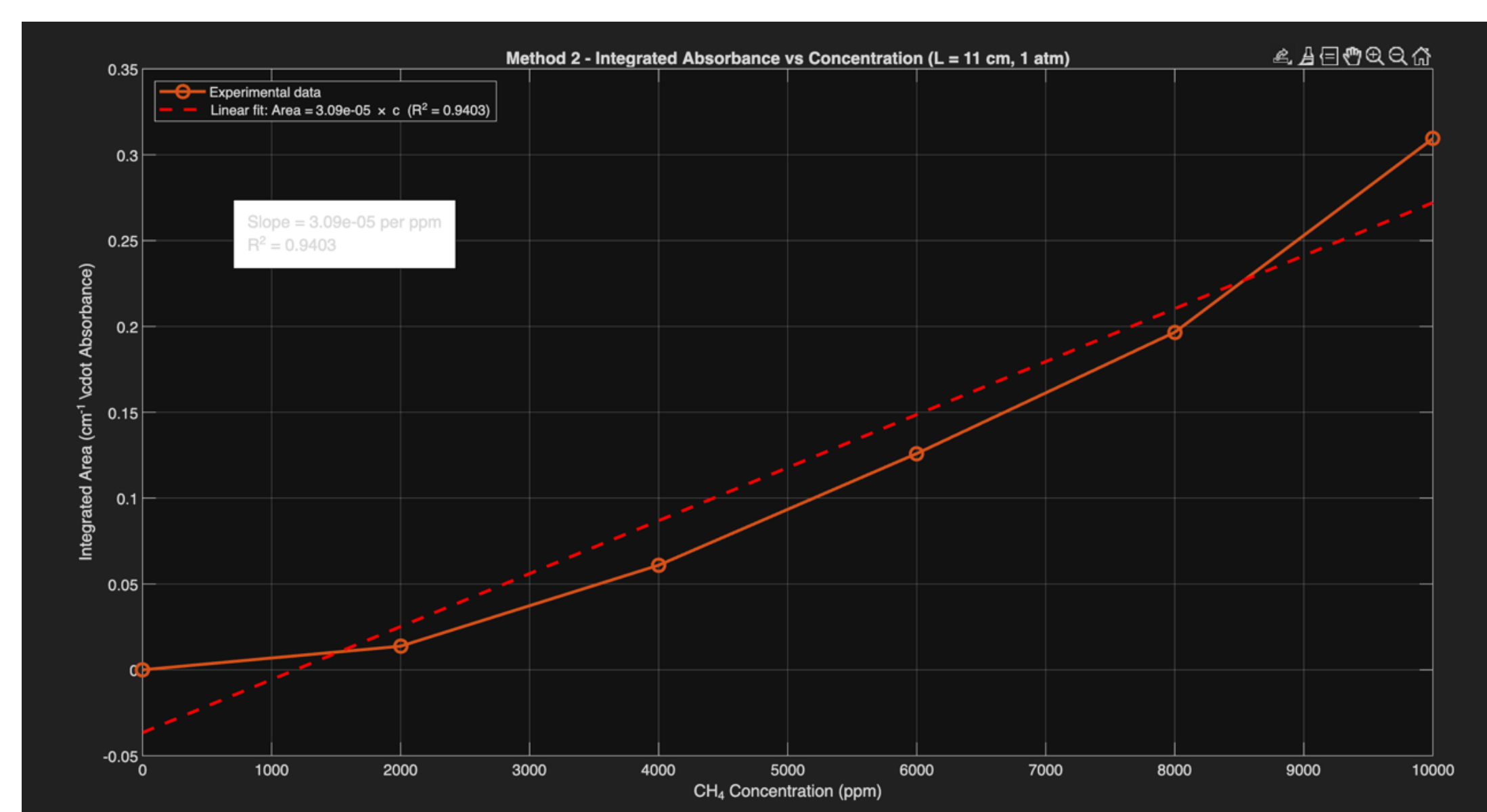
ii. Sub-section for the discussion

Figure 5. Absorbance calculation using Beer Law



Concentration	Beer-Lambert
0 ppm	0
2000 ppm	0.00230877
4000 ppm	0.00908884
6000 ppm	0.0171923
8000 ppm	0.0271742
10000 ppm	0.040186
R^2	0.9557

Figure 6. Absorbance calculation using Area difference method



Concentration	Area-Difference
0 ppm	0
2000 ppm	0.0137614
4000 ppm	0.0609371
6000 ppm	0.125966
8000 ppm	0.196626
10000 ppm	0.0309409
R^2	0.9403

- After the discussion, why the result / the point of 10000 ppm didn't show a linear form

1. Pressure Broadening

- Mole amount, and collision of methane in gas-cell increases, and increase the line width of the absorption curve and decrease the peak absorbance.

- Which will cause baseline fitting failure to be happen

2. Baseline Fitting Failure

- The baseline fitting failure will cause offset of the absorption line
- Peak absorbance will become smaller
- Which may cause the calculation error using both methods

IV. Conclusions and Future work

Conclude

1. Designed the short-range remote methane detector with optical path at 11cm and tested in ambient pressure.
2. Linearity R^2 is observed using 1. Beer law, 2. Area difference Method and confirmed with ≈ 0.95 , which is acceptable to be linearity of the curve.

Improvements can be made

1. Testing the system in vacuum environment, to have a more accurate result compared to the ambient situation
2. Add an extra calculation method to reduce the error from pressure broadening and the baseline fitting failure.

Limitations

1. The concentrations is only limited in 0-10000 ppm, and varies only for 2000 ppm
2. No temperature or pressure variations applied in this experiment.